

NAME: _____

PERIOD: _____

Overlap Does Not Decide Independence

AP Statistics · Unit 2 · Topic 2.7

[STAR] THE PROBLEM

Suppose a school surveys 200 students about whether they own at least one pet and whether they play a musical instrument. Let P be the event that a randomly selected surveyed student owns a pet and I the event that the student plays an instrument.

Aaliyah: *The events overlap because 84 students do both. Since they are not mutually exclusive, they must be independent.*

Marcus: *Overlap does not decide independence; the relevant probabilities must be compared.*

Pet ownership and instrument playing

Pet?	Plays	Does not	Total
Yes	84	36	120
No	16	64	80
Total	100	100	200

FIRST TAKE — one sentence before you work the parts

Does owning a pet seem to change the chance that a student plays an instrument? Give one table detail in one sentence.

[BOOK] INDEPENDENCE, INTERSECTION, AND UNION (keep on the page)

Events A and B are **independent** exactly when knowing one does not change the probability of the other: $P(A | B) = P(A)$. Equivalently, $P(A \cap B) = P(A)P(B)$. Establish independence before using that product. **Mutually exclusive** events have no overlap, so $P(A \cap B) = 0$; this is a different relationship. For any two events, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. To calculate a conditional probability from a table, divide the count in both events by the total for the given group.

Work (a)–(c).

DOK 1 (a) State the counts for P , I , and $P \cap I$.

DOK 2 (b) Find $P(I)$, $P(I | P)$, and $P(P \cup I)$. Show the fractions or setup.

DOK 3 (c) Whose reasoning is supported? Decide independence using the two instrument-playing probabilities. Explain why the 84 students in both groups settle mutual exclusivity but do not settle independence. Write 3–4 sentences.

Frame: The events are _____ because $P(I | P) = \underline{\hspace{1cm}}$ while $P(I) = \underline{\hspace{1cm}}$.

Frame: Mutually exclusive means _____, whereas independent means _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.